

Department of Statistics and Business Analytics
STAT 482 - Capstone in Analytics for Business

Spring 2026

Case Study: Multiple Regression in Biology & Analytics

Title: Modeling the Impact of Environmental Factors on Plant Growth Using Multiple Regression

Due submission date: Wednesday, April 1, 2026

Case Description

A biotechnology firm is investigating how environmental conditions affect plant growth rates in controlled ecosystems. The goal is to build a predictive model that explains variation in plant biomass based on several biological and environmental factors. Students will apply multiple regression techniques to analyze real-world biological data and generate actionable insights.

Objectives

By completing this case study, students will:

- Apply **multiple linear regression** to a real biological dataset
- Interpret relationships between predictors and plant growth
- Evaluate model assumptions (linearity, multicollinearity, heteroscedasticity, normality)
- Perform model selection and validation
- Translate statistical findings into **business and biological insights**

Data Source

Students will use an open biological dataset:

- **Dataset:** Plant Growth and Environmental Variables
- **Source:** UCI Machine Learning Repository
- **Link:** <https://archive.ics.uci.edu/ml/datasets/Forest+Fires>

(Alternative option if needed: <https://www.kaggle.com/datasets/uciml/forest-fires-dataset>)

Variables Definition

Dependent Variable (Response)

- **Burned Area (area)** – Continuous variable representing affected land (proxy for ecosystem impact/growth stress)

Independent Variables (Predictors)

- **Temperature (temp)** – °C

- **Relative Humidity (RH)** – %
- **Wind Speed (wind)** – km/h
- **Rainfall (rain)** – mm
- **FFMC (Fine Fuel Moisture Code)** – Moisture content indicator
- **DMC (Duff Moisture Code)** – Soil moisture indicator
- **DC (Drought Code)** – Long-term drought measure
- **ISI (Initial Spread Index)** – Fire spread potential

Recommended Solution Methodology

Step 1: Data Understanding & Cleaning

- Inspect missing values and outliers
- Apply transformations if needed (e.g., $\log(\text{area} + 1)$)
- Conduct exploratory data analysis (EDA)

Step 2: Model Specification

- Define a **multiple linear regression model**
- Consider transformations and interaction terms where appropriate

Step 3: Estimation Using R

Use the following R packages:

- **stats** – baseline regression modeling
- **car** – multicollinearity diagnostics (VIF)
- **ggplot2** – visualization
- **MASS** – stepwise model selection
- **caret** – model validation

Step 4: Diagnostic Testing

- Linearity check (residual plots)
- Multicollinearity (VIF)
- Normality (Q-Q plots)
- Homoscedasticity (Breusch-Pagan test)

Step 5: Model Selection

- Compare models using:
 - Adjusted R^2
 - AIC/BIC
- Perform **stepwise regression**

Step 6: Interpretation

- Interpret coefficients in biological context
 - Identify most influential environmental factors
 - Discuss managerial implications (e.g., agricultural planning, risk mitigation)
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Step 7: Validation

- Split data (train/test)
- Evaluate predictive performance (RMSE, MAE)

6. Deliverables

Students must submit:

- A **written case report**
- An **oral presentation** summarizing findings